N24 Quantum time correlation functions

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1

We introduce the notations in this first section which will be used throughout this note. The quantum time correlation function for two dynamical variables A and B is defined as

$$c_{AB}(t) = \langle A(0)B(t)\rangle \tag{1.1}$$

where

$$\langle \dots \rangle = \mathrm{tr}\left[\rho \cdots\right], \qquad \rho = \frac{1}{Q}e^{-\beta H}, \qquad Q = \mathrm{tr}\left[e^{-\beta H}\right] \tag{1.2}$$

$$B(\tau) = e^{iH\tau/\hbar}Be^{-i\hbar\tau/\hbar} \tag{1.3}$$

The Kubo-transformed time correlation function is defined as

$$\tilde{c}_{AB}(t) = \frac{1}{\beta}\int_0^\beta d\lambda \,\langle A(-i\hbar\lambda)B(t)\rangle \tag{1.4}$$

Notice that the ladder-operator correlator paper uses a different Fourier transform convention compared to the 2004 original RPMD paper. (they differ by a sign for the complex exponential) Let us use the same convention as 'Lattice thermal transport from phonon spectra beyond perturbation theory' paper here

$$f(t) = \frac{1}{2\pi}\int_{-\infty}^{\infty} d\omega \, e^{-i\omega t}F(\omega) \tag{1.5}$$

$$F(\omega) = \int_{-\infty}^{\infty} dt \, e^{i\omega t}f(t) \tag{1.6}$$

importantly, this will lead to a different relationship between the quantum time correlation function and and Kubo-transformed version (c.f. Eq. (15) in original RPMD paper)

$$\tilde{C}_{AB}(\omega) = \frac{\beta_\omega}{\beta}C_{AB}(\omega) \tag{1.7}$$

where effective reciprocal temperature is defined as

$$\beta_\omega = \frac{e^{\beta\hbar\omega} - 1}{\hbar\omega}. \tag{1.8}$$

which is so called 'harmonic correction factor' up to a Fourier transform convention.

2

We now aim to derive the leading order of error by using classical Boltzmann distribution in place of the quantum (or star, which will be defined later) Boltzmann distribution The Wigner transform of an operator A (see Appendix A)is defined as

$$\bar{A}(p,q) = \int dx \, \langle q - \frac{x}{2}|A|q + \frac{x}{2}\rangle e^{ixp/\hbar} \tag{2.1}$$

Note the following properties of the Wigner transform

1. $\overline{A+B} = \bar{A} + \bar{B}$
2. $\bar{A} = \bar{A}^*$ if $A = A^\dagger$
3. $\bar{1} = 1$
4. $\overline{q^n} = q'^n$
5. $\overline{p^n} = p'^n$

where q' and p' denotes the eigenvalue of position and momentum operators. The trace of the product of two operators have the following Wigner transform

$$\mathrm{tr}\,[AB] = \frac{1}{2\pi\hbar}\int dp \int dq \, \bar{A}\bar{B} \tag{2.2}$$

a particular useful case is the relationship between Hilbert space average and phase space average

$$\langle A\rangle = \mathrm{tr}\,[\rho A] = \frac{1}{2\pi\hbar}\int dp \int dq \, \bar{\rho}\bar{A} \tag{2.3}$$

$$\langle A\rangle = \frac{1}{2\pi\hbar Q}\int dp \int dq \, \overline{e^{-\beta H}}\bar{A} \tag{2.4}$$

The Wigner transform of a product between two operators is Wigner transform of a product of two operators is (proof: appendix A)

$$\overline{AB} = \bar{A} \star \bar{B} \tag{2.5}$$

where the star product $\star$ is defined as

$$\bar{A} \star \bar{B} \equiv \bar{A} e^{i\hbar\Lambda/2}\bar{B} \tag{2.6}$$

where

$$\Lambda \equiv \frac{\overleftarrow{\partial}}{\partial q}\frac{\overrightarrow{\partial}}{\partial p} - \frac{\overleftarrow{\partial}}{\partial p}\frac{\overrightarrow{\partial}}{\partial q} = \overleftrightarrow{\partial_q\partial_p} - \overleftrightarrow{\partial_p\partial_q} \tag{2.7}$$

the overhead arrows denotes the direction of operation of the differential operators

$$f\frac{\overleftarrow{\partial}}{\partial x}g \equiv \frac{\partial f}{\partial x}g \qquad f\frac{\overrightarrow{\partial}}{\partial x}g \equiv f\frac{\partial g}{\partial x} \tag{2.8}$$

hence

$$\begin{aligned}\overline{e^{-\beta H}} &= \overline{1-\beta H+\frac{\beta^2}{2!}H^2-\frac{\beta^3}{3!}H^3+\cdots} \\ &= \bar{1}-\beta\bar{H}+\frac{\beta^2}{2!}\overline{H^2}-\frac{\beta^2}{3!}\overline{H^3}+\cdots \\ &= 1-\beta\bar{H}+\frac{\beta^2}{2!}\bar{H}\star\bar{H}-\frac{\beta^3}{3!}\bar{H}\star\overline{H^2}+\cdots \\ &= 1-\beta\bar{H}+\frac{\beta^2}{2!}\bar{H}\star\bar{H}-\frac{\beta^3}{3!}\bar{H}\star\bar{H}\star\bar{H}+\cdots \end{aligned} \tag{2.9}$$

define the star power of a variable as

$$\bar{A}_\star^n \equiv \underbrace{\bar{A}\star\bar{A}\star\cdots\star\bar{A}}_{n \text{ times}}, \qquad \bar{A}_\star^0 = 1 \tag{2.10}$$

and define the star exponential as

$$e_\star^{\bar{A}} \equiv \sum_{n=0}^{\infty}\frac{\bar{A}_\star^n}{n!} \tag{2.11}$$

thus

$$\overline{e^{-\beta H}} = e_\star^{-\beta\bar{H}} \tag{2.12}$$

But since $\bar{H} = \mathcal{H}(p,q)$ (which follows from property 4 and 5 of the Wigner transform), we have

$$\overline{e^{-\beta H}} = e_\star^{-\beta\mathcal{H}(p,q)} \tag{2.13}$$

which is the classical Boltzmann distribution, except there is $\star$ at the bottom, we will compare quantum and classical statistics by look into the error caused by removing the $\star$. But we check partition function before proceeding

$$\begin{aligned} Q &= \frac{1}{2\pi\hbar}\int dp\int dq\, e_\star^{-\beta\mathcal{H}} \\ &= \frac{1}{2\pi\hbar}\int dp\int dq\int dx\,\langle q-\frac{x}{2}|e^{-\beta H}|q+\frac{x}{2}\rangle e^{ixp/\hbar} \\ &= \int dq\int dx\,\langle q-\frac{x}{2}|e^{-\beta H}|q+\frac{x}{2}\rangle\frac{1}{2\pi\hbar}\int dp\, e^{ixp/\hbar} \\ &= \int dq\int dx\,\langle q-\frac{x}{2}|e^{-\beta H}|q+\frac{x}{2}\rangle\delta(x) \\ &= \int dq\langle q|e^{-\beta H}|q\rangle \\ &= \mathrm{tr}\left[e^{-\beta H}\right] \end{aligned} \tag{2.14}$$

as expected. Consider

$$\mathcal{H}\star\mathcal{H} = \mathcal{H}e^{i\hbar\Lambda/2}\mathcal{H} = \mathcal{H}^2+\frac{i\hbar}{2}\mathcal{H}\Lambda\mathcal{H}+O(\hbar^2) \tag{2.15}$$

one can define the Poisson bracket $\{\cdot,\cdot\}$ via

$$\{\bar{A},\bar{B}\} \equiv \bar{A}\Lambda\bar{B} \tag{2.16}$$

the $\mathcal{H}\Lambda\mathcal{H}$ vanishes, therefore

$$\mathcal{H} \star \mathcal{H} = \mathcal{H}^2 + O(\hbar^2) \tag{2.17}$$

similarly

$$\begin{aligned}
\mathcal{H}_\star^3 &= \left(\mathcal{H}^2 + O(\hbar^2)\right) \star \mathcal{H} \\
&= (\mathcal{H}^2 + O(\hbar^2)) e^{i\hbar\Lambda/2}\mathcal{H} \\
&= (\mathcal{H}^2 + O(\hbar^2)) \left(1 + \frac{i\hbar}{2}\Lambda + O(\hbar^2)\right) \mathcal{H} \\
&= \mathcal{H}^3 + \frac{i\hbar}{2}\mathcal{H}^2\Lambda\mathcal{H} + O(\hbar^2)
\end{aligned} \tag{2.18}$$

it is easy to show, just like the commutator, the Poisson bracket obeys the product rule

$$\{fg, h\} = f\{g, h\} + \{g, h\}f \tag{2.19}$$

and it follows that

$$\{\mathcal{H}^2, \mathcal{H}\} = \mathcal{H}\{\mathcal{H}, \mathcal{H}\} + \{\mathcal{H}, \mathcal{H}\}\mathcal{H} = 0 \tag{2.20}$$

therefore

$$\mathcal{H}_\star^3 = \mathcal{H}^3 + O(\hbar^2) \tag{2.21}$$

then by induction

$$\mathcal{H}_\star^n = \mathcal{H}^n + O(\hbar^2) \tag{2.22}$$

and

$$\boxed{e_\star^{-\beta\mathcal{H}} = e^{-\beta\mathcal{H}} + O(\hbar^2)} \tag{2.23}$$

so replacing the quantum (star) Boltzmann distribution with the classical Boltzmann distribution introduces error of $O(\hbar^2)$. We can calculate this term more explicitly, consider

$$\Lambda^2 = \left(\overleftarrow{\partial_q}\overrightarrow{\partial_p} - \overleftarrow{\partial_p}\overrightarrow{\partial_q}\right)^2 = \overleftarrow{\partial_q^2}\overrightarrow{\partial_p^2} - 2\overleftarrow{\partial_q}\overrightarrow{\partial_p}\overleftarrow{\partial_p}\overrightarrow{\partial_q} + \overleftarrow{\partial_p^2}\overrightarrow{\partial_q^2} \tag{2.24}$$

hence

$$\mathcal{H}\Lambda^2\mathcal{H} = \mathcal{H}_{qq}\mathcal{H}_{pp} - 2\mathcal{H}_{qp}\mathcal{H}_{qp} + \mathcal{H}_{pp}\mathcal{H}_{qq} \tag{2.25}$$

since no term in $\mathcal{H}$ contain both q and p, the mixed second derivatives must vanish. Note $\mathcal{H}_{pp} = 1/m$ and $\mathcal{H}_{qq} = V''(q)$, we have

$$\mathcal{H}\Lambda^2\mathcal{H} = \frac{2}{m}V''(q) \tag{2.26}$$

similarly

$$\mathcal{H}\Lambda^3\mathcal{H} = \mathcal{H}\left(\overleftarrow{\partial_q}\overrightarrow{\partial_p} - \overleftarrow{\partial_p}\overrightarrow{\partial_q}\right)^3\mathcal{H} \tag{2.27}$$

Binomial expansion gives

$$\left(\overleftarrow{\partial_q}\overrightarrow{\partial_p} - \overleftarrow{\partial_p}\overrightarrow{\partial_q}\right)^3 = \overleftarrow{\partial_q^3}\overrightarrow{\partial_p^3} - 3\overleftarrow{\partial_q^2}\overrightarrow{\partial_p^2}\overleftarrow{\partial_p}\overrightarrow{\partial_q} + 3\overleftarrow{\partial_q}\overrightarrow{\partial_p}\overleftarrow{\partial_p^2}\overrightarrow{\partial_q^2} - \overleftarrow{\partial_p^3}\overrightarrow{\partial_q^3} \tag{2.28}$$

and

$$\mathcal{H}\Lambda^3\mathcal{H} = \mathcal{H}_{qqq}\mathcal{H}_{ppp} - 3\mathcal{H}_{qqp}\mathcal{H}_{ppq} + 3\mathcal{H}_{qpp}\mathcal{H}_{pqq} - \mathcal{H}_{ppp}\mathcal{H}_{qqq} \tag{2.29}$$

all mixed 3rd derivative vanishes along with $\mathcal{H}_{ppp}$ therefore

$$\mathcal{H}\Lambda^3\mathcal{H} = 0 \tag{2.30}$$

and for higher order derivative it is clear that even the non-mixed derivative terms such as $\mathcal{H}_{qqqq}\mathcal{H}_{pppp} = 0$ (for Λ^4) since the kinetic part of the Hamiltonian only goes up to p^2, therefore

$$\boxed{\mathcal{H}\Lambda^n\mathcal{H} = \frac{2}{m}V''(q)\delta_{n2}} \tag{2.31}$$

so

$$\boxed{\mathcal{H}\star\mathcal{H} = \mathcal{H}^2 - \frac{\hbar^2}{4m}V''(q)} \tag{2.32}$$

which is exact. Define a local harmonic frequency $\omega(q)$ via

$$m\omega^2(q) \equiv V''(q) \tag{2.33}$$

thus

$$\boxed{\mathcal{H}\star\mathcal{H} = \mathcal{H}^2 - \left(\frac{\hbar\omega(q)}{2}\right)^2} \tag{2.34}$$

define a local zero-point energy

$$E_0(q) \equiv \frac{1}{2}\hbar\omega(q) \tag{2.35}$$

the star product between two $\mathcal{H}$ looks quite clean

$$\boxed{\mathcal{H}\star\mathcal{H} = \mathcal{H}^2 - E_0^2(q) = (\mathcal{H} + E_0(q))(\mathcal{H} - E_0(q))} \tag{2.36}$$

if $E_0(q) <<< \mathcal{H}$, then we hit the classical limit and simply remove the $\star$ between two $\mathcal{H}$. But for exact quantum case, it seems like the star product is a the product of two fictitious classical Hamiltonians: one with potential shifted up by local zero-point energy and the other one shifted down by local zero-point energy. To derive the explicit $O(\hbar^2)$ leading quantum correction to classical Boltzmann distribution, consider the expansion up to $O(\hbar^2)$

$$e_\star^{-\beta\mathcal{H}} = e^{-\beta\mathcal{H}}\left(1 + \hbar^2 w_2\right) \tag{2.37}$$

where $w_2(p, q, \beta)$ is to be determined. Consider

$$\frac{\partial}{\partial\beta}e^{-\beta H} = -He^{-\beta H} \tag{2.38}$$

taking the Wigner tranform of both sides gives

$$\frac{\partial}{\partial\beta}e_\star^{-\beta\mathcal{H}} = -\mathcal{H}\star e^{-\beta\mathcal{H}} \tag{2.39}$$

substitute in the previous expansion of star Boltzmann distribution, then compare $O(\hbar^0)$ gives

$$\frac{\partial}{\partial\beta}e^{-\beta\mathcal{H}} = -\mathcal{H}e^{-\beta\mathcal{H}} \tag{2.40}$$

as expected. It is also simple to show by comparing $O(\hbar^2)$ gives

$$\frac{\partial w_2}{\partial\beta} = \frac{1}{8}e^{\beta\mathcal{H}}\mathcal{H}\Lambda^2 e^{-\beta\mathcal{H}} \tag{2.41}$$

and from Eq. (2.26) that

$$\mathcal{H}\Lambda^2 e^{-\beta\mathcal{H}} = V''(q)\partial_p^2 e^{-\beta\mathcal{H}} + \frac{1}{m}\partial_q^2 e^{-\beta\mathcal{H}} \tag{2.42}$$

thus

$$\frac{\partial w_2}{\partial\beta} = \frac{\beta^2}{8}V''(q)\left(\frac{\partial\mathcal{H}}{\partial p}\right)^2 - \frac{\beta}{8}V''(q)\frac{\partial^2\mathcal{H}}{\partial p^2} + \frac{\beta^2}{8m}\left(\frac{\partial\mathcal{H}}{\partial q}\right)^2 - \frac{\beta}{8m}\frac{\partial^2\mathcal{H}}{\partial q^2} \tag{2.43}$$

$$\frac{\partial w_2}{\partial\beta} = \frac{\beta^2}{8}V''(q)\left(\frac{p}{m}\right)^2 - \frac{\beta}{4m}V''(q) + \frac{\beta^2}{8m}\left[V'(q)\right]^2 \tag{2.44}$$

integrate with respect to β, and note $w(\beta = 0)$ since the quantum correction vanishes at classical limit, we have

$$w_2 = \frac{\beta^3}{24}V''(q)\left(\frac{p}{m}\right)^2 - \frac{\beta^2}{8m}V''(q) + \frac{\beta^3}{24m}\left[V'(q)\right]^2 \tag{2.45}$$

so the leading correction to classical Boltzmann distribution is

$$\boxed{w_2 = \frac{\beta^3}{24m}\left(\frac{p^2}{m^2}V''(q) + \frac{[V'(q)]^2}{m}\right) - \frac{\beta^2}{8m}V''(q)} \tag{2.46}$$

which agrees with the original results obtained by E. Wigner. (Appendix C)

3

Now we move onto dynamics and look for the first quantum correction term to classical dynamics.

The Heisenberg equation of motion for A is

$$i\hbar\frac{\partial A}{\partial t} = [A, H] \tag{3.1}$$

$$\frac{\partial A}{\partial t} = \frac{1}{i\hbar}[A, H] \tag{3.2}$$

for Wigner transforms, we simply put a bar over everything

$$\frac{\partial\bar{A}}{\partial t} = \frac{1}{i\hbar}\overline{[A, H]} \tag{3.3}$$

we proceed by defining the Moyal bracket

$$\{\bar{A}, \bar{B}\}_\star \equiv \frac{1}{i\hbar}\{\bar{A}\star\bar{B} - \bar{B}\star\bar{A}\} \tag{3.4}$$

note the following relationship with the commutator

$$\frac{1}{i\hbar}\overline{[A, B]} = \{\bar{A}, \bar{B}\}_\star \tag{3.5}$$

also note (proof: appendix A)

$$\{\bar{A}, \bar{B}\}_\star = \frac{2}{\hbar}\bar{A}\sin\left(\frac{\hbar}{2}\Lambda\right)\bar{B} \tag{3.6}$$

clearly

$$\lim_{\hbar\to 0}\{\bar{A},\bar{B}\}_\star = \{\bar{A},\bar{B}\} \tag{3.7}$$

and also

$$\{\bar{A},\bar{B}\}_\star = \{\bar{A},\bar{B}\} + O(\hbar^2) \tag{3.8}$$

thus

$$\frac{\partial \bar{A}}{\partial t} = \{\bar{A},\mathcal{H}\}_\star = \frac{2}{\hbar}\bar{A}\sin\left(\frac{\hbar}{2}\Lambda\right)\mathcal{H} \tag{3.9}$$

Again, knocking this star off recovers classical Hamiltonian dynamics, along with knocking off the star in the $\star$-Boltzmann distribution, quantum statistical mechanics would be reduced to classical statistical mechanics.

$$\bar{A}\Lambda^3\mathcal{H} = \bar{A}_{qqq}\mathcal{H}_{ppp} - 3\bar{A}_{qqp}\mathcal{H}_{ppq} + 3\bar{A}_{qpp}\mathcal{H}_{pqq} - \bar{A}_{ppp}\mathcal{H}_{qqq} \tag{3.10}$$

for Hamiltonian of the form

$$\mathcal{H}(p,q) = \frac{p^2}{2m} + V(q) \tag{3.11}$$

since no term in $\mathcal{H}$ contain both p and q, we have $\mathcal{H}_{ppq} = \mathcal{H}_{pqq} = 0$ and also $\mathcal{H}_{ppp} = 0$, the only non-zero Hamiltonian derivative is $\mathcal{H}_{qqq} = V_{qqq} \equiv V^{(3)}$. The leading order correction to classical Hamiltonian dynamics is

$$\boxed{\frac{\partial \bar{A}}{\partial t} = \{\bar{A},\mathcal{H}\} + \frac{\hbar^2}{24}\frac{\partial^3 \bar{A}}{\partial p^3}V^{(3)} + O(\hbar^4)} \tag{3.12}$$

which clearly vanishes for the special case of harmonic potential, but in general is non-zero. We rewrite the equation above as

$$\frac{\partial \bar{A}}{\partial t} = \mathcal{L}_0\bar{A} + \hbar^2\mathcal{L}_2\bar{A} + O(\hbar^4) \tag{3.13}$$

Expand the symbol up to $O(\hbar^2)$

$$\bar{A}(t) = \bar{A}_0(t) + \hbar^2 D_2\bar{A}(t) + O(\hbar^4) \tag{3.14}$$

where D_2 shall be determined. Substitute this back into equations of motion

$$\frac{\partial \bar{A}_0}{\partial t} + \hbar^2\frac{\partial D_2\bar{A}(t)}{\partial t} = \mathcal{L}_0\bar{A}_0 + \hbar^2\left(\mathcal{L}_0 D_2\bar{A}(t) + \mathcal{L}_2\bar{A}_0\right) \tag{3.15}$$

comparing $O(\hbar^0)$ terms on both sides gives the classical equations of motion

$$\frac{\partial \bar{A}_0}{\partial t} = \mathcal{L}_0\bar{A}_0 \tag{3.16}$$

which has the solution

$$B_0(t) = e^{\mathcal{L}_0 t}A_0(0) \tag{3.17}$$

comparing $O(\hbar^2)$ terms on both sides gives the leading order of quantum correction to the dynamics

$$\frac{\partial D_2\bar{A}(t)}{\partial t} = \mathcal{L}_0 D_2\bar{A}(t) + \mathcal{L}_2\bar{A}_0(t) \tag{3.18}$$

$$\frac{\partial D_2\bar{A}(t)}{\partial t} - \mathcal{L}_0 D_2\bar{A}(t) = \mathcal{L}_2\bar{A}_0(t) \tag{3.19}$$

this is clearly solvable by an integrating factor of $e^{-t\mathcal{L}_0}$

$$e^{-t\mathcal{L}_0}\frac{\partial D_2\bar{A}(t)}{\partial t} - e^{-t\mathcal{L}_0}\mathcal{L}_0 D_2\bar{A}(t) = e^{-t\mathcal{L}_0}\mathcal{L}_2\bar{A}_0(t) \tag{3.20}$$

$$\frac{\partial}{\partial t}\left(e^{-t\mathcal{L}_0}D_2\bar{A}(t)\right) = e^{-t\mathcal{L}_0}\mathcal{L}_2\bar{A}_0(t) \tag{3.21}$$

using the initial condition that $\bar{A}(0) = \bar{A}_0$, integrating both sides gives

$$e^{-t\mathcal{L}_0}D_2\bar{A}(t) = \int_0^t dt'\, e^{-t'\mathcal{L}_0}\mathcal{L}_2\bar{A}_0(t') \tag{3.22}$$

$$\boxed{D_2\bar{A}(t) = \int_0^t dt'\, e^{(t-t')\mathcal{L}_0}\mathcal{L}_2\bar{A}_0(t')} \tag{3.23}$$

and we know everything on RHS from classical dynamics. Therefore,

$$\boxed{\bar{A}(t) = \bar{A}_0(t) + \hbar^2\int_0^t dt'\, e^{(t-t')\mathcal{L}_0}\mathcal{L}_2\bar{A}(t')} \tag{3.24}$$

4

We now have both the correction for classical (Boltzmann) statistics from section 2 and correction for classical (Hamiltonian) dynamics in section 3, we now return to the Kubo-transformed time correlation function in terms of Wigner transform

$$\begin{aligned}
\tilde{c}_{AB}(t) &= \frac{1}{\beta}\int_0^\beta d\lambda\, \mathrm{tr}[e^{-(\beta-\lambda)H}Ae^{-\lambda H}B(t)] \\
&= \frac{1}{\beta}\int_0^\beta d\lambda \int dp \int dq\, \overline{e^{-(\beta-\lambda)H}Ae^{-\lambda H}}\bar{B}(p,q,t) \\
&= \frac{1}{\beta}\int_0^\beta d\lambda \int dp \int dq\, \overline{e^{-(\beta-\lambda)H}} \star \bar{A} \star \overline{e^{-\lambda H}}\bar{B}(p,q,t) \\
&= \frac{1}{\beta}\int_0^\beta d\lambda \int dp \int dq\, e_\star^{-(\beta-\lambda)\mathcal{H}} \star \bar{A} \star e_\star^{-\lambda\mathcal{H}}\bar{B}(p,q,t)
\end{aligned} \tag{4.1}$$

define Kubo-transformed symbol

$$\tilde{A}_\beta = \frac{1}{\beta}\int_0^\infty d\lambda\, e_\star^{-(\beta-\lambda)\mathcal{H}} \star \bar{A} \star e_\star^{-\lambda\mathcal{H}} \tag{4.2}$$

so

$$\tilde{c}_{AB}(t) = \frac{1}{2\pi\hbar}\int dp \int dq\, \tilde{A}_\beta\bar{B}(p,q,t) \tag{4.3}$$

and

$$
\begin{aligned}
e_\star^{-(\beta-\lambda)\mathcal{H}} \star \bar{A} \star e_\star^{-\lambda\mathcal{H}} &= e^{-(\beta-\lambda)\mathcal{H}} \star \bar{A} \star e^{-\lambda\mathcal{H}} + O(\hbar^2) \\
&= e^{-(\beta-\lambda)\mathcal{H}} e^{i\hbar\Lambda/2} \bar{A} e^{i\hbar\Lambda/2} e^{-\lambda\mathcal{H}} + O(\hbar^2) \\
&= e^{-(\beta-\lambda)\mathcal{H}} \left(1 + \frac{i\hbar}{2}\Lambda\right) \bar{A} \left(1 + \frac{i\hbar}{2}\Lambda\right) e^{-\lambda\mathcal{H}} + O(\hbar^2) \\
&= e^{-\beta\mathcal{H}} \bar{A} + \frac{i\hbar}{2} \{e^{-(\beta-\lambda)\mathcal{H}}, \bar{A}\} e^{-\lambda\mathcal{H}} + \frac{i\hbar}{2} e^{-(\beta-\lambda)\mathcal{H}} \{\bar{A}, e^{-\lambda\mathcal{H}}\} + O(\hbar^2) \\
&= e^{-\beta\mathcal{H}} \bar{A} + \frac{i\hbar}{2} \left(-(\beta-\lambda) e^{-(\beta-\lambda)\mathcal{H}} \{\mathcal{H}, \bar{A}\} e^{-\lambda\mathcal{H}} + e^{-(\beta-\lambda)\mathcal{H}} \{\bar{A}, \mathcal{H}\} e^{-\lambda\mathcal{H}} (-\lambda) + O(\hbar^2)\right) \\
&= e^{-\beta\mathcal{H}} \bar{A} + \frac{i\hbar}{2} \left((2\lambda - \beta) e^{-\beta\mathcal{H}} \{\mathcal{H}, \bar{A}\}\right) + O(\hbar^2) \qquad (4.4)
\end{aligned}
$$

and obviously the integral over λ kills the $O(\hbar)$ terms.

So when the Star Boltzmann distribution is smeared between two positions in front of and behind $\bar{A}$, the lower order correction is $O(\hbar^2)$ from this part of the integral. However, the $O(\hbar)$ is in general non-vanishing for the ordinary $e^{-(\beta-\lambda)\mathcal{H}} \star \bar{A}$, this is perhaps another reason why the Kubo-transformed time correlation function $\tilde{c}_{AB}(t)$ is the more classical object compared to the normal $c_{AB}(t)$ along with all the symmetries of $\tilde{c}_{AB}(t)$. Define $\mathcal{K}_2$ via the expansion up to $O(\hbar^2)$

$$
\frac{1}{\beta} \int_0^\beta d\lambda \, e_\star^{-(\beta-\lambda)\mathcal{H}} \star \bar{A} \star e_\star^{-\lambda\mathcal{H}} = e^{-\beta\mathcal{H}} \bar{A} \left(1 + \hbar^2 \mathcal{K}\right) \qquad (4.5)
$$

to find this $\mathcal{K}_2$, define

$$
\mathcal{H}_\eta = \mathcal{H} + \eta \bar{A} \qquad (4.6)
$$

and note the following familiar identity

$$
-\frac{1}{\beta} \frac{\partial}{\partial\eta} e_\star^{-\beta\mathcal{H}_\eta} \bigg|_{\eta=0} = \frac{1}{\beta} \int_0^\beta d\lambda \, e_\star^{-(\beta-\lambda)\mathcal{H}} \star \bar{A} \star e_\star^{-\lambda\mathcal{H}}. \qquad (4.7)
$$

Now use

$$
e_\star^{-\beta\mathcal{H}_\eta} = e^{-\beta\mathcal{H}_\eta} \left[1 + \hbar^2 w_2[\mathcal{H}_\eta] + O(\hbar^4)\right]. \qquad (4.8)
$$

Differentiating,

$$
\begin{aligned}
-\frac{1}{\beta} \frac{\partial}{\partial\eta} e_\star^{-\beta\mathcal{H}_\eta} \bigg|_{\eta=0} &= -\frac{1}{\beta} \frac{\partial}{\partial\eta} \left\{e^{-\beta\mathcal{H}_\eta} \left[1 + \hbar^2 w_2[\mathcal{H}_\eta]\right]\right\} \bigg|_{\eta=0} + O(\hbar^4) \\
&= -\frac{1}{\beta} e^{-\beta\mathcal{H}} \left\{-\beta\bar{A} \left[1 + \hbar^2 w_2[\mathcal{H}]\right] + \hbar^2 \frac{\partial w_2[\mathcal{H}_\eta]}{\partial\eta} \bigg|_{\eta=0}\right\} + O(\hbar^4) \\
&= e^{-\beta\mathcal{H}} \left[\bar{A} + \hbar^2 \left(\bar{A} w_2[\mathcal{H}] - \frac{1}{\beta} \frac{\partial w_2[\mathcal{H} + \eta\bar{A}]}{\partial\eta} \bigg|_{\eta=0}\right)\right] + O(\hbar^4). \qquad (4.9)
\end{aligned}
$$

Therefore,

$$
\boxed{\mathcal{K}_2 = \bar{A} w_2[\mathcal{H}] - \frac{1}{\beta} \frac{\partial w_2[\mathcal{H} + \eta\bar{A}]}{\partial\eta} \bigg|_{\eta=0}} \qquad (4.10)
$$

So there are two leading corrections to classical time correlation functions, the first one is a $O(\hbar^2)$ correction to classical statistics

$$\boxed{c_{AB}^{(s)}(t) = \frac{\hbar^2}{2\pi\hbar}\int dp \int dq\, \mathcal{K}_2 e^{-\beta\mathcal{H}} \bar{A}\bar{B}_0(p_t, q_t)} \tag{4.11}$$

and a second correction term of equal order $O(\hbar^2)$ to classical dynamics

$$\boxed{c_{AB}^{(d)}(t) = \frac{\hbar^2}{2\pi\hbar}\int dp \int dq\, e^{-\beta\mathcal{H}} \bar{A} \int_0^t dt'\, e^{(t-t')\mathcal{L}_0} \mathcal{L}_2 \bar{B}_0(p_{t'}, q_{t'})} \tag{4.12}$$

Appendix A

The Wigner transform of an operator A is defined as

$$\bar{A}(p,q) \equiv \int dx \langle q - \frac{x}{2}|A|q + \frac{x}{2}\rangle e^{ixp/\hbar} \tag{A1}$$

insert two complete sets of momenta

$$\begin{aligned}
\bar{A}(p,q) &= \int dx \int dp' \int dp''\, \langle q - \frac{x}{2}|p'\rangle\langle p'|A|p''\rangle\langle p''|q + \frac{x}{2}\rangle e^{ixp/\hbar} \\
&= \frac{1}{2\pi\hbar}\int dx \int dp' \int dp''\, e^{i(q-x/2)p'/\hbar}\langle p'|A|p''\rangle e^{-i(q+x/2)p''/\hbar} e^{ixp/\hbar} \\
&= \frac{1}{2\pi\hbar}\int dx \int dp' \int dp''\, \langle p'|A|p''\rangle e^{iq(p'-p'')/\hbar} e^{-ix(p'+p'')/2\hbar} e^{ixp/\hbar} \\
&= \frac{1}{2\pi\hbar}\int dx \int dp' \int dp''\, \langle p'|A|p''\rangle e^{iq(p'-p'')/\hbar} e^{ix(p-(p'+p'')/2)/\hbar} \\
&= \int dp' \int dp''\, \langle p'|A|p''\rangle e^{iq(p'-p'')/\hbar} \delta\left(p - \frac{p'+p''}{2}\right)
\end{aligned} \tag{A2}$$

define

$$P = \frac{p'+p''}{2}, \qquad y = p' - p'' \tag{A3}$$

so that

$$p' = P + \frac{y}{2}, \qquad p'' = P - \frac{y}{2}. \tag{A4}$$

The Jacobian is

$$\begin{aligned}
\left|\frac{\partial(p',p'')}{\partial(P,y)}\right| &= \left|\det\begin{pmatrix} \frac{\partial p'}{\partial P} & \frac{\partial p'}{\partial y} \\ \frac{\partial p''}{\partial P} & \frac{\partial p''}{\partial y} \end{pmatrix}\right| \\
&= \left|\det\begin{pmatrix} 1 & \frac{1}{2} \\ 1 & -\frac{1}{2} \end{pmatrix}\right| \\
&= \left|-\frac{1}{2} - \frac{1}{2}\right| = 1.
\end{aligned} \tag{A5}$$

therefore

$$\begin{aligned}\bar{A} &= \int dP \int dy \, \langle P + \frac{y}{2}|A|P - \frac{y}{2}\rangle e^{iqy/\hbar} \delta\left(p - P\right) \\ &= \int dy \, \langle p + \frac{y}{2}|A|p - \frac{y}{2}\rangle e^{iqy/\hbar}\end{aligned} \tag{A6}$$

making the substitution $x' = -x$ in Eq. (A1) then removing the prime mark gives

$$\bar{A} = \int dx \langle q + \frac{x}{2}|A|q - \frac{x}{2}\rangle e^{-ixp/\hbar} \tag{A7}$$

and similarly making the substitution $y' = -y$ in Eq. (A4) then removing the prime mark after gives

$$\bar{A} = \int dy \langle p - \frac{y}{2}|A|p + \frac{y}{2}\rangle e^{-iqy/\hbar} \tag{A8}$$

For Hermitian operator $A = A^\dagger$, taking the complex conjugate of both sides of Eq. (A1) then use Eq. (A7) gives

$$\bar{A}^* = \int dx \, \langle q + \frac{x}{2}|A|q - \frac{x}{2}\rangle e^{-ixp/\hbar} = \bar{A} \tag{A9}$$

thus the Wigner transform of a Hermitian operator is real. Now let $A = q$,

$$\begin{aligned}\bar{q} &= \int dx \, \langle q' - \frac{x}{2}|q|q' + \frac{x}{2}\rangle e^{ixp/\hbar} \\ &= \int dx \, \langle q' - \frac{x}{2}|q' + \frac{x}{2}\rangle \left(q' + \frac{x}{2}\right) e^{ixp/\hbar} \\ &= \int dx \, \delta\left(q' + \frac{x}{2} - q' + \frac{x}{2}\right) \left(q' + \frac{x}{2}\right) e^{ixp/\hbar} \\ &= \int dx \, \delta\left(x\right) \left(q' + \frac{x}{2}\right) e^{ixp/\hbar} \\ &= q'\end{aligned} \tag{A10}$$

so the Wigner transform of the position operator is just its eigenvalue. It also immediately follows that

$$\overline{q^n} = q'^n \tag{A11}$$

for an operator that can be expressed as a power series of q, such as potential energy

$$\overline{V(q)} = V(q') \tag{A12}$$

similarly

$$\overline{p^n} = p'^n \tag{A13}$$

and the kinetic energy

$$\frac{1}{2m}\overline{p^2} = \frac{1}{2m}p'^2 \tag{A14}$$

and since the Wigner transform is linear,

$$\overline{H(p,q)} = \frac{1}{2m}\overline{p^2} + \overline{V(q)} = \frac{1}{2m}p'^2 + V(q') = \mathcal{H}(p',q') \tag{A15}$$

so the Wigner transform of quantum Hamiltonian operator is the classical Hamiltonian. Furthermore, the Wigner transform of the unit operator is

$$\bar{1} = 1 \tag{A16}$$

Now consider the phase space integral

$$\begin{aligned}\frac{1}{2\pi\hbar}\int dp \int dq\, \bar{A}\bar{B} &= \frac{1}{2\pi\hbar}\int dp \int dq \int dx\, \langle q - \frac{x}{2}|A|q + \frac{x}{2}\rangle e^{ixp/\hbar} \int dx'\, \langle q + \frac{x'}{2}|B|q - \frac{x'}{2}\rangle e^{-ix'p/\hbar} \\ &= \int dq \int dx \langle q - \frac{x}{2}|A|q + \frac{x}{2}\rangle \int dx'\, \langle q + \frac{x'}{2}|B|q - \frac{x'}{2}\rangle \delta(x - x') \\ &= \int dq \int dx \langle q - \frac{x}{2}|A|q + \frac{x}{2}\rangle\, \langle q + \frac{x}{2}|B|q - \frac{x}{2}\rangle \end{aligned} \tag{A17}$$

define

$$q' \equiv q - \frac{x}{2}, \qquad q'' \equiv q + \frac{x}{2} \tag{A18}$$

the Jacobian is 1, similar to Eq. (A5), therefore

$$\frac{1}{2\pi\hbar}\int dp \int dq\, \bar{A}\bar{B} = \int dq' \int dq'' \langle q'|A|q''\rangle\, \langle q''|B|q'\rangle = \mathrm{tr}\,[AB] \tag{A19}$$

This is an important result: The trace of product of two operators in Hilbert space is the same as the phase space integral of the product of the Wigner transforms. For example that quantum average of an observable is the trace against the density operator ρ

$$\boxed{\langle A\rangle = \mathrm{tr}\,[\rho A] = \frac{1}{2\pi\hbar}\int dp \int dq\, \bar{\rho}\bar{A}} \tag{A20}$$

Wigner transform of a product of two operators

$$\begin{aligned}\overline{AB} &= \int dx\, \langle q - \frac{x}{2}|AB|q + \frac{x}{2}\rangle e^{ixp/\hbar} \\ &= \int dx \int dy\, \langle q - \frac{x}{2}|A|y\rangle\langle y|B|q + \frac{x}{2}\rangle e^{ixp/\hbar} \end{aligned} \tag{A21}$$

proceed with substitutions

$$a = y - q + \frac{x}{2}, \qquad b = q + \frac{x}{2} - y \tag{A22}$$

therefore

$$x = a + b, \qquad y = q + \frac{a - b}{2} \tag{A23}$$

the Jacobian is again 1, therefore Eq. (A21) can be rewritten as

$$\begin{aligned}\overline{AB} &= \int da \int db\, \langle q - \frac{a+b}{2}|A|q + \frac{a-b}{2}\rangle\langle q + \frac{a-b}{2}|B|q + \frac{a+b}{2}\rangle e^{iap/\hbar} e^{ibp/\hbar} \\ &= \int da \int db\, \langle q - \frac{a}{2} - \frac{b}{2}|A|q + \frac{a}{2} - \frac{b}{2}\rangle\langle q + \frac{a}{2} - \frac{b}{2}|B|q + \frac{a}{2} + \frac{b}{2}\rangle e^{iap/\hbar} e^{ibp/\hbar} \\ &= \int da \int db\, \langle q - \frac{b}{2} - \frac{a}{2}|A|q - \frac{b}{2} + \frac{a}{2}\rangle\langle q + \frac{a}{2} - \frac{b}{2}|B|q + \frac{a}{2} + \frac{b}{2}\rangle e^{iap/\hbar} e^{ibp/\hbar} \end{aligned} \tag{A24}$$

note that

$$\bar{A}(p, q-\frac{b}{2}) = \int da\, \langle q-\frac{b}{2}-\frac{a}{2}|A|q-\frac{b}{2}+\frac{a}{2}\rangle e^{iap/\hbar} \tag{A25}$$

the inverse transform is

$$\langle q-\frac{b}{2}-\frac{a}{2}|A|q-\frac{b}{2}+\frac{a}{2}\rangle = \frac{1}{2\pi\hbar}\int dp'\, \bar{A}(p', q-\frac{b}{2})e^{-iap'/\hbar} \tag{A26}$$

and similarly for B. Substitute Eq. (A26) back into Eq. (A24) yields

$$\begin{aligned}\overline{AB} &= \frac{1}{(2\pi\hbar)^2}\int da\int db\int dp'\int dp''\, \bar{A}(p', q-\frac{b}{2})e^{-iap'/\hbar}\bar{B}(p'', q+\frac{a}{2})e^{-ibp''/\hbar}e^{iap/\hbar}e^{ibp/\hbar} \\ &= \frac{1}{(2\pi\hbar)^2}\int da\int db\int dp'\int dp''\, \bar{A}(p', q-\frac{b}{2})\bar{B}(p'', q+\frac{a}{2})e^{ia(p-p')/\hbar}e^{ib(p-p'')/\hbar}\end{aligned} \tag{A27}$$

To proceed, notice that for an arbitrary function $f(x)$

$$f(x+a) = e^{a\frac{\partial}{\partial x}}f(x) \tag{A28}$$

where

$$e^{a\frac{\partial}{\partial x}} = \sum_{n=1}^{\infty}\frac{a^n}{n!}\frac{\partial^n}{\partial x^n} \tag{A29}$$

and clearly

$$\bar{A}(p', q-\frac{b}{2}) = e^{-\frac{b}{2}\frac{\partial}{\partial q}}\bar{A}(p', q) \tag{A30}$$

and

$$\bar{B}(p'', q+\frac{a}{2}) = e^{\frac{a}{2}\frac{\partial}{\partial q}}\bar{B}(p'', q) \tag{A31}$$

substitute these results back into Eq. (A27) gives

$$\overline{AB} = \frac{1}{(2\pi\hbar)^2}\int da\int db\int dp'\int dp''\, \bar{A}(p', q)e^{-\frac{b}{2}\frac{\overleftarrow{\partial}}{\partial q}}e^{\frac{a}{2}\frac{\overrightarrow{\partial}}{\partial q}}\bar{B}(p'', q)e^{ia(p-p')/\hbar}e^{ib(p-p'')/\hbar} \tag{A32}$$

where the overhead arrows denotes the direction of operation of the differential operators

$$f\frac{\overleftarrow{\partial}}{\partial x}g \equiv \frac{\partial f}{\partial x}g \qquad f\frac{\overrightarrow{\partial}}{\partial x}g \equiv f\frac{\partial g}{\partial x} \tag{A33}$$

$$\overline{AB} = \frac{1}{(2\pi\hbar)^2}\sum_{nm}\int da\int db\int dp'\int dp''\, \bar{A}(p', q)\left(-\frac{b}{2}\right)^n\frac{1}{n!}\frac{\overleftarrow{\partial^n}}{\partial q^n}\frac{\overrightarrow{\partial^m}}{\partial q^m}\frac{1}{m!}\left(\frac{a}{2}\right)^m \bar{B}(p'', q)e^{ia(p-p')/\hbar}e^{ib(p-p'')/\hbar} \tag{A34}$$

$$\overline{AB} = \frac{1}{(2\pi\hbar)^2}\sum_{nm}\int da\int db\int dp'\int dp''\, \bar{A}(p', q)e^{ib(p-p'')/\hbar}b^n\left(-\frac{1}{2}\right)^n\frac{1}{n!}\frac{\overleftarrow{\partial^n}}{\partial q^n}\frac{\overrightarrow{\partial^m}}{\partial q^m}\frac{1}{m!}\left(\frac{1}{2}\right)^m \bar{B}(p'', q)a^m e^{ia(p-p')/\hbar} \tag{A35}$$

to proceed, consider

$$\delta(p-p') = \frac{1}{2\pi\hbar}\int dx\, e^{ix(p-p')/\hbar} \tag{A36}$$

$$\frac{\partial^n}{\partial p^n}\delta(p-p') = \frac{1}{2\pi\hbar}\int dx\,\left(\frac{ix}{\hbar}\right)^n e^{ix(p-p')/\hbar} \tag{A37}$$

$$(-i\hbar)^n\frac{\partial^n}{\partial p^n}\delta(p-p') = \frac{1}{2\pi\hbar}\int dx\,x^n e^{ix(p-p')/\hbar} \tag{A38}$$

return to Eq. (A35)

$$\overline{AB} = \sum_{nm}\int dp'\int dp''\,\bar{A}(p',q)(-i\hbar)^n\frac{\partial^n\delta(p-p'')}{\partial p^n}\left(-\frac{1}{2}\right)^n\frac{1}{n!}\overleftarrow{\frac{\partial^n}{\partial q^n}}\overrightarrow{\frac{\partial^m}{\partial q^m}}\frac{1}{m!}\left(\frac{1}{2}\right)^m(-i\hbar)^m\frac{\partial^m\delta(p-p')}{\partial p^m}\bar{B}(p'',q) \tag{A39}$$

$$\overline{AB} = \sum_{nm}\int dp''\,\frac{\partial^m\bar{A}(p,q)}{\partial p^m}(-i\hbar)^n\frac{\partial^n\delta(p-p'')}{\partial p^n}\left(-\frac{1}{2}\right)^n\frac{1}{n!}\overleftarrow{\frac{\partial^n}{\partial q^n}}\overrightarrow{\frac{\partial^m}{\partial q^m}}\frac{1}{m!}\left(\frac{1}{2}\right)^m(-i\hbar)^m\bar{B}(p'',q) \tag{A40}$$

$$\overline{AB} = \sum_{nm}\frac{\partial^m\bar{A}(p,q)}{\partial p^m}(-i\hbar)^n\left(-\frac{1}{2}\right)^n\frac{1}{n!}\overleftarrow{\frac{\partial^n}{\partial q^n}}\overrightarrow{\frac{\partial^m}{\partial q^m}}\frac{1}{m!}\left(\frac{1}{2}\right)^m(-i\hbar)^m\frac{\partial^n\bar{B}(p,q)}{\partial p^n} \tag{A41}$$

$$\overline{AB} = \sum_{nm}\bar{A}(p,q)(-i\hbar)^n\left(-\frac{1}{2}\right)^n\frac{1}{n!}\overleftarrow{\frac{\partial^n}{\partial q^n}}\overleftarrow{\frac{\partial^m}{\partial p^m}}\overrightarrow{\frac{\partial^n}{\partial p^n}}\overrightarrow{\frac{\partial^m}{\partial q^m}}\frac{1}{m!}\left(\frac{1}{2}\right)^m(-i\hbar)^m\bar{B}(p,q) \tag{A42}$$

$$\overline{AB} = \bar{A}(p,q)\sum_{nm}\left(\frac{i\hbar}{2}\right)^n\frac{1}{n!}\overleftarrow{\frac{\partial^n}{\partial q^n}}\overrightarrow{\frac{\partial^n}{\partial p^n}}\overleftarrow{\frac{\partial^m}{\partial p^m}}\overrightarrow{\frac{\partial^m}{\partial q^m}}\frac{1}{m!}\left(-\frac{i\hbar}{2}\right)^m\bar{B}(p,q) \tag{A43}$$

$$\overline{AB} = \bar{A}(p,q)e^{\frac{i\hbar}{2}\overleftarrow{\frac{\partial}{\partial q}}\overrightarrow{\frac{\partial}{\partial p}}}e^{-\frac{i\hbar}{2}\overleftarrow{\frac{\partial}{\partial p}}\overrightarrow{\frac{\partial}{\partial q}}}\bar{B}(p,q) \tag{A44}$$

the two exponential operators clearly commutes, therefore through all that mess in the previous 3 pages, we end up with something quite neat

$$\overline{AB} = \bar{A}(p,q)e^{\frac{i\hbar}{2}\left(\overleftarrow{\frac{\partial}{\partial q}}\overrightarrow{\frac{\partial}{\partial p}}-\overleftarrow{\frac{\partial}{\partial p}}\overrightarrow{\frac{\partial}{\partial q}}\right)}\bar{B}(p,q) \tag{A45}$$

define

$$\Lambda \equiv \overleftarrow{\frac{\partial}{\partial q}}\overrightarrow{\frac{\partial}{\partial p}} - \overleftarrow{\frac{\partial}{\partial p}}\overrightarrow{\frac{\partial}{\partial q}} \tag{A46}$$

define the star product $\star$

$$\bar{A}\star\bar{B} \equiv \bar{A}e^{i\hbar\Lambda/2}\bar{B} \tag{A47}$$

therefore

$$\boxed{\overline{AB} = \bar{A}\star\bar{B}} \tag{A48}$$

thus the Wigner transform of a commutator is

$$\overline{[A,B]} = \overline{AB} - \overline{BA} = \bar{A}\star\bar{B} - \bar{B}\star\bar{A} \tag{A49}$$

define the Moyal bracket

$$\{\bar{A},\bar{B}\}_\star \equiv \frac{1}{i\hbar}\left(\bar{A}\star\bar{B} - \bar{B}\star\bar{A}\right) \tag{A50}$$

therefore

$$\boxed{\frac{1}{i\hbar}\overline{[A,B]} = \{\bar{A},\bar{B}\}_\star} \tag{A51}$$

We now make a connection with Poisson bracket, for two classical dynamical variables f and g

$$\{f,g\} \equiv \frac{\partial f}{\partial q}\frac{\partial g}{\partial p} - \frac{\partial f}{\partial p}\frac{\partial g}{\partial q} \tag{A52}$$

using directional derivative operators, it can also be written as

$$\{f,g\} = f\left(\frac{\overleftarrow{\partial}}{\partial q}\frac{\overrightarrow{\partial}}{\partial p} - \frac{\overleftarrow{\partial}}{\partial p}\frac{\overrightarrow{\partial}}{\partial q}\right)g = f\Lambda g \tag{A53}$$

analogous to the commutator, the Poisson bracket is also anti-symmetric upon exchanging two of its arguments

$$\{f,g\} = -\{g,f\} \tag{A54}$$

equivalently

$$f\Lambda g = -g\Lambda f \tag{A55}$$

so swapping $f \leftrightarrow g$ is same as $\Lambda \to -\Lambda$, thus

$$g\Lambda^n f = (-1)^n f\Lambda^n g \tag{A56}$$

and

$$\begin{aligned}
\bar{B}e^{i\hbar\Lambda/2}\bar{A} &= \sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{i\hbar}{2}\right)^n \bar{B}\Lambda^n\bar{A} \\
&= \sum_{n=0}^{\infty}\frac{1}{n!}\left(\frac{i\hbar}{2}\right)^n (-1)^n\bar{A}\Lambda^n\bar{B} \\
&= \bar{A}e^{-i\hbar\Lambda/2}\bar{B}
\end{aligned} \tag{A57}$$

therefore

$$\begin{aligned}
\{\bar{A},\bar{B}\}_\star &= \frac{1}{i\hbar}\left(\bar{A}\star\bar{B} - \bar{B}\star\bar{A}\right) \\
&= \frac{1}{i\hbar}\left(\bar{A}e^{i\hbar\Lambda/2}\bar{B} - \bar{B}e^{i\hbar\Lambda/2}\bar{A}\right) \\
&= \frac{1}{i\hbar}\left(\bar{A}e^{i\hbar\Lambda/2}\bar{B} - \bar{A}e^{-i\hbar\Lambda/2}\bar{B}\right) \\
&= \frac{1}{i\hbar}\bar{A}\left(e^{i\hbar\Lambda/2} - e^{-i\hbar\Lambda/2}\right)\bar{B} \\
&= \frac{2}{\hbar}\bar{A}\sin\left(\frac{\hbar}{2}\Lambda\right)\bar{B}
\end{aligned} \tag{A58}$$

this is another neat but important result

$$\boxed{\{\bar{A},\bar{B}\}_\star = \frac{2}{\hbar}\bar{A}\sin\left(\frac{\hbar}{2}\Lambda\right)\bar{B}} \tag{A59}$$

and the classical limit is recovered from $\sin x \simeq x$,

$$\boxed{\{\bar{A},\bar{B}\} = \lim_{\hbar\to 0}\{\bar{A},\bar{B}\}_\star} \tag{A60}$$

so taking the limit of $\hbar \to 0$ is same as knocking the $\star$ off. It is also clear that

$$\{\bar{A},\bar{B}\}_\star = \{\bar{A},\bar{B}\} + O(\hbar^2) \tag{A61}$$

Appendix B

The Feynman-Kleinert smeared potential is

$$V_{a^2}(q) \equiv \frac{1}{\sqrt{2\pi a^2}}\int dq'\, e^{-(q-q')^2/2a^2} V(q') \tag{B1}$$

make the substitution $\xi = q' - q$ so $q' = q + \xi$ and

$$\begin{aligned} V_{a^2}(q) &= \frac{1}{\sqrt{2\pi a^2}}\int d\xi\, e^{-\xi^2/2a^2} V(q+\xi) \\ &= \frac{1}{\sqrt{2\pi a^2}}\int d\xi\, e^{-\xi^2/2a^2} e^{\xi\partial_q} V(q) \\ &= \frac{1}{\sqrt{2\pi a^2}}\sum_{n=0}^{\infty}\int d\xi\, e^{-\xi^2/2a^2}\frac{\xi^n}{n!}\frac{\partial^n}{\partial q^n}V(q) \\ &= \frac{1}{\sqrt{2\pi a^2}}\sum_{n=0}^{\infty}\frac{1}{n!}\frac{\partial^n}{\partial q^n}V(q)\int d\xi\, e^{-\xi^2/2a^2}\xi^n \end{aligned} \tag{B2}$$

note the integral over ξ vanishes for odd n by parity, and also note the following standard result

$$\int_{-\infty}^{\infty} dx\, x^{2n} e^{-\alpha x^2} = \sqrt{\frac{\pi}{\alpha}}\frac{(2n-1)!!}{(2\alpha)^n} \tag{B3}$$

where the double factor means for example $7!! = 7\times 5\times 3\times 1$. Therefore

$$\frac{1}{\sqrt{2\pi a^2}}\int d\xi\, e^{-\xi^2/2a^2}\xi^{2m} = (2m-1)!! a^{2m} \tag{B4}$$

so

$$V_{a^2}(q) = \sum_{m=0}^{\infty}\frac{1}{(2m)!}\frac{\partial^{2m}}{\partial q^{2m}}V(q)(2m-1)!! a^{2m} \tag{B5}$$

then note the following relationship

$$(2n)! = (2n)!!(2n-1)!! = 2^n n!(2n-1)!! \tag{B6}$$

thus

$$\frac{(2n-1)!!}{(2n)!} = \frac{1}{2^n n!} \tag{B7}$$

therefore

$$V_{a^2}(q) = \sum_{m=0}^{\infty}\frac{a^{2m}}{2^m m!}\frac{\partial^{2m}}{\partial q^{2m}}V(q) = \sum_{m=0}^{\infty}\frac{1}{m!}\left(\frac{a^2}{2}\frac{\partial^2}{\partial q^2}\right)^m V(q) \tag{B8}$$

finally the smeared potential is

$$\boxed{V_{a^2}(q) = e^{a^2\partial_q^2/2}V(q)} \tag{B9}$$

Appendix C

Wigner himself has painstakingly obtained the following results

$$e_{\star}^{-\beta\mathcal{H}} = \Omega(p,q) = \sum_{n=0}^{\infty} \hbar^{2n} \Omega_n(p,q). \tag{C1}$$

$$\Omega_0(p,q) = e^{-\beta\mathcal{H}} \tag{C2}$$

and the higher order corrections to classical Boltzmann distribution

$$\Omega_1(p,q) = \Omega_0(p,q) \left\{ \frac{1}{8} V''(q) \left[-\frac{\beta^2}{m} + \frac{\beta^3 p^2}{3m^2} \right] + \frac{\beta^3 [V'(q)]^2}{24m} \right\}. \tag{C3}$$

$$\begin{aligned}
\Omega_2(p,q) = & \frac{\beta^2 \Omega_0(p,q)}{16m^2} \left(\frac{\beta^2 p^4}{m^2} - \frac{6\beta p^2}{m} + 3 \right) \left(\frac{\beta^2 [V''(q)]^2}{72} - \frac{\beta V^{(4)}(q)}{120} \right) \\
& + \frac{\beta^2 \Omega_0(p,q)}{64m^2} \left(\frac{2\beta p^2}{m} - 2 \right) \left(\frac{\beta^3 V''(q)[V'(q)]^2}{18} - \frac{2\beta^2 [V''(q)]^2}{15} - \frac{\beta^2 V'(q) V^{(3)}(q)}{15} + \frac{\beta V^{(4)}(q)}{15} \right) \\
& + \frac{\beta^3 \Omega_0(p,q)}{64m^2} \left(\frac{\beta^3 [V'(q)]^4}{18} - \frac{22\beta^2 V''(q)[V'(q)]^2}{45} + \frac{2\beta [V''(q)]^2}{5} + \frac{8\beta V'(q) V^{(3)}(q)}{15} - \frac{4V^{(4)}(q)}{15} \right)
\end{aligned} \tag{C4}$$

it would be interesting to see how this maps onto the centroid distribution of the ring polymer.